

L13 SU Candidate Leg C27 — Composite v0.4 Substrate-Mass-Mechanism at Gen-2

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L13 SU Candidate Leg C27 v0.1

Claim

$$m_{\mu}/m_e = n_C \cdot (n_C/\text{rank})_3 + N_c^4/2^{N_c} = \frac{1575}{8} + \frac{81}{8} = \frac{1656}{8} = 207$$

substrate-natural integer at 0.112% Tier 1/Tier 2 boundary precision vs observed $m_{\mu}/m_e \approx 206.768$.

Independent Substrate-Mechanism Support

1. Term 1 ($n_C \cdot (5/2)_3 = 196.875$): *substrate-Bergman kernel exponent $n_C/\text{rank}=5/2$ (Saturday May 28 One-Genus convention) + substrate-Pochhammer Schur II $\text{Sym}^3 V_{\text{fund}} \text{Sp}(2)$ substrate-K-type $V(3/2, 1/2)$ substrate-natural rising-factorial form*
2. Term 2 ($N_c^4 / 2^{N_c} = 10.125$): *substrate-color N_c raised to substrate-Casey-#14-codim-4 substrate-natural exponent + substrate-Cartan-base 2 raised to substrate-color N_c substrate-natural per Casey #14 STANDING substrate-codim-4 cross-link*

Cross-Domain Failure if Substrate-Mechanism Absent

If substrate-Bergman kernel exponent were not $n_C/\text{rank}=5/2$ (e.g., if genus were $g=7$ not $n_C=5$), substrate-Pochhammer would give $(7/2)_3 = 693/8$ substantive WRONG form. Saturday May 28 One-Genus convention substrate-natural FORCING form.

If Casey #14 STANDING substrate-codim-4 were absent, N_c^4 substrate-natural exponent would not be substrate-mechanism-forced. Casey #14 STANDING substrate-codim-4 substrate-mechanism class FORCING.

Strong-Uniqueness Contribution

Adds 1 leg to v1.6 STANDALONE 11 legs $\rightarrow$ 13 if ratified (with C26).

Cumulative null-model: $(1/3)^{13} \approx 6.3 \times 10^{-7}$.

Multi-Week RIGOROUS Promotion Path

- explicit FK Ch. XII §VI substrate-Bergman matrix element at $V(3/2, 1/2)$
- explicit Casey #13 per-generation substrate-mechanism class $N_c^{\text{codim-4}}$ derivation
- explicit Casey #14 STANDING substrate-codim-4 cascade derivation
- substantive composite sum substrate-natural form match at Tier 1/Tier 2 boundary

— Lyra, Thu 2026-06-04 11:24 EDT. L13 SU candidate leg C27 v0.1: Composite v0.4 substrate-mass-mechanism at gen-2 $V(3/2, 1/2)$ substrate-natural form 207 at 0.112%; SU 13 if ratified ($6.3e-7$ null-model).